

Roshan Mohan

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Engineer with 9 years of experience in designing, testing, and manufacturing of cutting-edge hardware technologies.

EXPERIENCE

Senior Hardware Test Engineer

February 2024- Present

Tesla

Palo Alto, California

- Developing and implementing test programs for electric powertrain technologies for autonomous vehicles.
- Performing component, system and vehicle validation under dynamic and environmental conditions.
- Investigating root cause of failures to drive product improvements.
- Interfacing and simulating complex vehicular systems to validate firmware-hardware interactions.
- Mentoring junior engineers and technicians on engineering tasks.

Senior Mechanical Design Engineer

Apr 2019 - Jun 2023

Nunam

Bangalore, India

A technology company backed by Audi, pioneering advancements in lithium-ion battery energy storage.

- Led a cross-functional team to deliver the mechanical design, analysis, and production of three advanced lithium-ion battery products for clients including Audi AG and TATA Motors.
- Managed mechanical design from concept to production, including solid modeling, failure analysis and testing to ensure high reliability. Extensively utilized additive manufacturing techniques for prototype builds.
- Drove technical maturation and testing activities, significantly improving designs and reducing production costs through DFMA.
- Designed mechanical subsystems meeting UL and AIS safety standards for shock and thermal performance.

Mechanical Design Engineer

Oct 2017 - Apr 2019

Biodesign Innovation Labs

Bangalore, India

A med-tech startup supported by the Government of India.

- Spearheaded the mechanical design of Respiraid, a Class 3 critical care medical device, now in use in 500+ ICUs.
- Led design reviews to ensure reliability, ensuring adherence to rigorous safety standards.
- Co-authored the patent for Respiraid's electro-mechanical drive design (USPO No. 20200261672).
- Developed a cost-effective CPAP breathing circuit specifically for pediatric pneumonia treatment, used in a controlled clinical study of 50 patients.

Design Engineer

Feb 2017 - Sep 2017

Nunam

Bangalore, India

- Designed and prototyped a 500Wh solar-powered battery for off-grid use based on extensive field research in Eastern India.

EDUCATION

University of California Berkeley, College of Engineering

Berkeley, California

Master of Design in Engineering

Dec 2024

- GPA 3.83
- Masters Thesis: Rapid Volumetric Additive Manufacturing for Sustainable Consumer Electronics
- Graduate Student Instructor for the Department of Nuclear Engineering (ENGIN 125)

Mahatma Gandhi University

Kerala, India

Bachelor of Mechanical Engineering

May 2016

- Outstanding Performance Award, 2016

SKILLS

3D Design- SolidWorks, Fusion 360, CATIA, NX.

Analysis- Ansys, SolidWorks Simulation, DFM, DFMEA.

Prototyping- 3D Printing: FDM, SLS, SLA, VAM, Silicone Casting, Laser Cutting,

Programming- Python, html & C++.

Manufacturing- CNC Machining, Sheet-metal, Carpentry, Metalworking, Injection Molding- Soft and Hard Tooling, Casting.